United States Patent	12408298
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Chae; Jae Min et al.

Vehicular coolant system and method of controlling the same

Abstract

A coolant system includes a battery coolant line fluidly connected to a battery, a power electronics coolant line fluidly connected to a power electronic component, a radiator coolant line fluidly connected to a radiator, a control valve configured to allow the battery coolant line, the power electronics coolant line, and the radiator coolant line to be fluidly connected to or separated from each other, a temperature sensor disposed in at least one of the battery coolant line, the power electronics coolant line, or the radiator coolant line, and a controller configured to control a switching operation of the control valve under a predetermined condition in a manner that fluidly connects a coolant line without any temperature sensor to a coolant line with the temperature sensor, and determine a temperature of a coolant passing through the coolant line without any temperature sensor based on a temperature sensed by the temperature sensor.

Inventors: **Chae; Jae Min (Seoul, KR), Ryu; Chang Gi (Seoul, KR), Lee; Soo II (Hwaseong-si, KR), Song; Dae Hyun (Seoul, KR), Lee; Seung Ryeol (Hwaseong-si, KR), Gwak; Jong Dae (Anyang-si, KR)**

Applicant: **HYUNDAI MOTOR COMPANY** (Seoul, KR); **KIA CORPORATION** (Seoul, KR)

Family ID: **94259496**

Assignee: **Hyundai Motor Company (Seoul, KR); Kia Corporation (Seoul, KR)**

Appl. No.: **18/488716**

Filed: **October 17, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20250031345 A1	Jan. 23, 2025

Foreign Application Priority Data

KR	10-2023-0092615	Jul. 17, 2023

Publication Classification

Int. Cl.: **H05K7/20** (20060101); **B60H1/00** (20060101); **B60L58/26** (20190101); **H01M10/48** (20060101); **H01M10/613** (20140101); **H01M10/625** (20140101); **H01M10/635** (20140101); **H01M10/6568** (20140101); **H01M10/663** (20140101)

U.S. Cl.:

CPC **H05K7/20281** (20130101); **B60H1/00278** (20130101); **B60L58/26** (20190201); **H01M10/486** (20130101); **H01M10/613** (20150401); **H01M10/625** (20150401); **H01M10/635** (20150401); **H01M10/6568** (20150401); **H01M10/663** (20150401); **H05K7/20263** (20130101); **H05K7/20272** (20130101); **H05K7/20927** (20130101); B60H2001/00307 (20130101); B60H1/00921 (20130101); B60H2001/00928 (20130101); B60L2240/545 (20130101); H01M2220/20 (20130101)

Field of Classification Search

CPC: H05K (7/20281); H05K (7/20263); H05K (7/20272); H05K (7/20927); H01M (10/613); H01M (10/625); H01M (10/635); H01M (10/6568); H01M (10/663); H01M (10/486); H01M (2220/20); B60L (58/26); B60L (2240/545); B60H (1/00278); B60H (1/00921); B60H (2001/00307); B60H (2001/00928)

USPC: 361/699; 700/299; 700/300

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4415847	12/1982	Galloway	165/300	B60L 53/302
6329723	12/2000	Hilpert	123/41.12	B60L 9/00
6394210	12/2001	Matsuda	903/903	B60H 1/004
6481230	12/2001	Kimishima	62/238.7	B60L 53/11
7451808	12/2007	Busse	62/238.7	F01P 3/20
10259291	12/2018	Aldridge	N/A	B60L 58/27
11331979	12/2021	Jeong	N/A	B60H 1/00278
11659696	12/2022	McEathron	361/679.47	H05K 7/20881
2019/0039439	12/2018	Aldridge	N/A	B60L 58/26
2020/0070619	12/2019	Kim	N/A	B60H 1/00921
2022/0161630	12/2021	Lindquist	N/A	B60L 1/003
2022/0181722	12/2021	Kim	N/A	B60K 11/02
2022/0355645	12/2021	Jeong	N/A	B60H 1/00664
2023/0104670	12/2022	Mothier	429/437	H01M 8/04037
2023/0158908	12/2022	Hou	320/109	B60L 3/0046
2024/0353188	12/2023	Reddy	N/A	H01M 10/6569

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2012022507	12/2011	WO	F01P 7/165

Primary Examiner: Vortman; Anatoly

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims the benefit of priority to Korean Patent Application No. 10-2023-0092615, filed on Jul. 17, 2023, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a vehicular coolant system and a method of controlling the same, and more particularly, to a vehicular coolant system and a method of controlling the same designed to improve electric efficiency as well as achieving cost savings by reducing the number of temperature sensors provided in an integrated coolant system for maintaining a power electronic component and/or a battery at appropriate temperatures.

BACKGROUND

(3) With a growing interest in energy efficiency and environmental issues, there is a demand for development of eco-friendly vehicles that can replace internal combustion engine vehicles. Such eco-friendly vehicles are classified into electric vehicles which are driven using fuel cells or electricity as a power source and hybrid vehicles which are driven using an engine and a battery.

(4) Electric vehicles or hybrid vehicles may include a vehicular thermal management system for air conditioning in a passenger compartment and maintaining a battery and/or a power electronic component at optimal temperatures. The vehicular thermal management system may include a heating, ventilation, and air conditioning (HVAC) system for air conditioning in the passenger compartment, and an integrated coolant system for maintaining the power electronic component of a power electronics system and/or the battery at appropriate temperatures.

(5) The integrated coolant system may include a battery coolant line fluidly connected to the battery, a power electronics coolant line fluidly connected to the power electronic component, and a radiator coolant line fluidly connected to a radiator. The battery coolant line may be thermally connected to the HVAC system through a battery chiller, and the power electronics coolant line may be thermally connected to the HVAC system through a heat exchanger. The radiator may be disposed adjacent to a front grille of the vehicle.

(6) In addition, the integrated coolant system may include a control valve controlling the flow of a coolant between the battery coolant line, the power electronics coolant line, and the radiator coolant line. The battery coolant line, the power electronics coolant line, and the radiator coolant line may be fluidly connected or separated by the control valve, and accordingly the coolant may be allowed to selectively circulate in the battery coolant line, the power electronics coolant line, and/or the radiator coolant line.

(7) The integrated coolant system according to the related art may include a radiator-side temperature sensor provided in the radiator coolant line, and a battery-side temperature sensor provided in the battery coolant line. The radiator-side temperature sensor may sense the temperature of the coolant passing through the radiator coolant line, and the battery-side temperature sensor may sense the temperature of the coolant passing through the battery coolant line. In addition, the temperature of the coolant passing through the power electronics coolant line may be indirectly estimated using a temperature sensor attached to the power electronic component. For example, the power electronic component may be a motor, an inverter, and the like. The inverter may be configured to control the motor, and a heat sink may be attached to a

circuit board of the inverter. The power electronics coolant line may be fluidly connected to the heat sink of the inverter, and the temperature sensor may be mounted on the heat sink of the inverter and sense the temperature of the heat sink. When the vehicle is traveling at a constant speed, the temperature of the coolant passing through the power electronics coolant line becomes similar to a temperature sensed by the temperature sensor mounted on the heat sink with an error of about 1° C. Accordingly, the temperature of the coolant passing through the power electronics coolant line may be determined based on the temperature sensed by the temperature sensor mounted on the heat sink.

(8) In the event of a rapid acceleration or deceleration of the vehicle (i.e., rapid changes in vehicle speed or in motor torque), the temperature of the heat sink rapidly changes in response to the changes in vehicle speed, but the temperature of the coolant passing through the power electronics coolant line does not rapidly change in response to the changes in vehicle speed. Accordingly, during the rapid acceleration or deceleration of the vehicle, the temperature sensed by the temperature sensor mounted on the heat sink may considerably differ from the temperature of the coolant passing through the power electronics coolant line. In order to accurately sense the temperature of the coolant passing through the power electronics coolant line, an additional temperature sensor may be provided in the power electronics coolant line. That is, three temperature sensors may be provided in the battery coolant line, the power electronics coolant line, and the radiator coolant line, respectively. As the three temperature sensors are provided, the overall manufacturing cost may increase.

SUMMARY

(9) An aspect of the present disclosure provides a vehicular coolant system and a method of controlling the same designed to improve electric efficiency as well as achieving cost savings by reducing the number of temperature sensors provided in an integrated coolant system for maintaining a power electronic component and/or a battery at appropriate temperatures.

(10) According to an aspect of the present disclosure, a vehicular coolant system may include: a battery coolant line fluidly connected to a battery; a power electronics coolant line fluidly connected to a power electronic component; a radiator coolant line fluidly connected to a radiator; a control valve configured to allow the battery coolant line, the power electronics coolant line, and the radiator coolant line to be fluidly connected to or separated from one another; a temperature sensor disposed at at least one of the battery coolant line, the power electronics coolant line, or the radiator coolant line; and a controller configured to control the control valve to: fluidly connect (i) a first coolant line that does not have the temperature sensor among the battery coolant line, the power electronics coolant line, and the radiator coolant line to (ii) a second coolant line that has the temperature sensor among the battery coolant line, the power electronics coolant line, and the radiator coolant line, and determine a first temperature of a coolant passing through the first coolant line based on a second temperature sensed by the temperature sensor disposed at the second coolant line.

(11) The temperature sensor may be disposed at the battery coolant line.

(12) The coolant system may comprise (i) a first temperature sensor disposed at the battery coolant line and (ii) a second temperature sensor disposed at the radiator coolant line.

(13) The coolant system may further include a heat sink temperature sensor disposed at a heat sink of the power electronic component.

(14) The control valve may comprise: a first battery-side port connected to an inlet of the battery coolant line; a second battery-side port connected to an outlet of the battery coolant line; a first power electronic-side port connected to an inlet of the power electronics coolant line; a second power electronic-side port connected to an outlet of the power electronics coolant line; a first radiator-side port connected to an inlet of the radiator coolant line; and a second radiator-side port connected to an outlet of the radiator coolant line.

(15) The control valve may be configured to allow two or more of the first battery-side port, the

second battery-side port, the first power electronic-side port, the second power electronic-side port, the first radiator-side port, or the second radiator-side port to selectively communicate with one another.
(16) The coolant system may further include a battery pump fluidly connected to the battery coolant line.
(17) The coolant system may further include a battery chiller that is fluidly connected to the battery coolant line and configured to thermally connect the battery coolant line to a heating, ventilation, and air conditioning (HVAC) system.
(18) The coolant system may further include a power electronic pump fluidly connected to the power electronics coolant line.
(19) The coolant system may further comprise a heat exchanger that is fluidly connected to the power electronics coolant line. The heat exchanger may be configured to thermally connect the power electronics coolant line to an HVAC system.
(20) According to another aspect of the present disclosure, a method for controlling a coolant system including a battery coolant line fluidly connected to a battery, a power electronics coolant line fluidly connected to a power electronic component, a radiator coolant line fluidly connected to a radiator, and a control valve configured to control flow of a coolant through the battery coolant line, the power electronics coolant line, and the radiator coolant line may comprise: obtaining a temperature of a heat sink of the power electronic component; determining whether a variation of the temperature of the heat sink of the power electronic component is greater than or equal to a predetermined threshold in a first circulation mode in which the power electronics coolant line is fluidly separated from the battery coolant line; based on determining that the variation in the temperature of the heat sink of the power electronic component is greater than or equal to the predetermined threshold, switching to a second circulation mode in which the power electronics coolant line is fluidly connected to the battery coolant line; obtaining a first temperature sensed by a temperature sensor that is disposed at the battery coolant line; and determining a second temperature of the coolant passing through the power electronics coolant line based on the first temperature sensed by the temperature sensor disposed at the battery coolant line.
(21) The method may further comprise: based on determining that the variation of the temperature of the heat sink of the power electronic component is less than the predetermined threshold, obtaining a third temperature that is sensed by a heat sink temperature sensor; and determining the second temperature of the coolant passing through the power electronics coolant line based on the third temperature sensed by the heat sink temperature sensor.
(22) According to another aspect of the present disclosure, a method for controlling a coolant system including a battery coolant line fluidly connected to a battery, a power electronics coolant line fluidly connected to a power electronic component, a radiator coolant line fluidly connected to a radiator, and a control valve configured to control flow of a coolant through the battery coolant line, the power electronics coolant line, and the radiator coolant line may include: obtaining a temperature of a heat sink of the power electronic component; determining whether a variation of the temperature of the heat sink of the power electronic component is greater than or equal to a first predetermined threshold in a first circulation mode in which the power electronics coolant line is fluidly separated from the battery coolant line and the radiator coolant line; based on determining that the variation of the temperature of the heat sink of the power electronic component is greater than or equal to the first predetermined threshold, determining a temperature difference between a first temperature of the coolant passing through the battery coolant line and the temperature of the heat sink; determining whether the temperature difference is greater than or equal to a second predetermined threshold; based on determining that the temperature difference is greater than or equal to the second predetermined threshold, switching to a second circulation mode in which the power electronics coolant line is fluidly connected to the radiator coolant line; and obtaining a first sensor temperature sensed by a first temperature sensor disposed at the radiator coolant line; and

determining a second temperature of the coolant passing through the power electronics coolant line based on the first sensor temperature sensed by the first temperature sensor disposed at the radiator coolant line.
(23) The method may further include obtaining a second sensor temperature sensed by a heat sink temperature sensor based on determining that the variation of the temperature of the heat sink of the power electronic component is less than the first predetermined threshold; determining the second temperature of the coolant passing through the power electronics coolant line based on the second sensor temperature sensed by the heat sink temperature sensor.
(24) The method may further include: switching to a third circulation mode in which the power electronics coolant line is fluidly connected to the battery coolant line based on determining that the temperature difference is less than the second predetermined threshold; obtaining a second sensor temperature sensed by a second temperature sensor disposed at the battery coolant line; and determining the second temperature of the coolant passing through the power electronics coolant line based on the second sensor temperature sensed by a temperature sensor disposed at the battery coolant line.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates an example of a vehicular thermal management system including a coolant system.
(2) FIG. **2** illustrates an example of a vehicular thermal management system including a coolant system.
(3) FIG. **3** illustrates an example of the flow of a coolant circulating in a first circulation mode by a first switching operation of a control valve in a coolant system.
(4) FIG. **4** illustrates an example of the flow of a coolant circulating in a second circulation mode by a second switching operation of a control valve in a coolant system.
(5) FIG. **5** illustrates an example of the flow of a coolant circulating in a third circulation mode by a third switching operation of a control valve in a coolant system.
(6) FIG. **6** illustrates an example of the flow of a coolant circulating in a fourth circulation mode by a fourth switching operation of a control valve in a coolant system.
(7) FIG. **7** illustrates an example of a flowchart of a method of controlling a coolant system.
(8) FIG. **8** illustrates an example of a flowchart of a method of controlling a coolant system.
DETAILED DESCRIPTION
(9) Hereinafter, exemplary implementations of the present disclosure will be described in detail with reference to the accompanying drawings. In the drawings, the same reference numerals will be used throughout to designate the same or equivalent elements. In addition, a detailed description of well-known techniques associated with the present disclosure will be ruled out in order not to unnecessarily obscure the gist of the present disclosure.
(10) Referring to FIG. **1**, a vehicular thermal management system may include a coolant system **10** configured to maintain a battery **21** and power electronic components **25***a* and **25***b* at appropriate temperatures, and a heating, ventilation, and air conditioning (HVAC) system **40** thermally connected to the coolant system **10**.
(11) In some implementations, the coolant system **10** may include a battery coolant line **11** fluidly connected to the battery **21**, a power electronics coolant line **12** fluidly connected to the power electronic components **25***a* and **25***b*, and a radiator coolant line **13** fluidly connected to a radiator **27**.
(12) In some examples, the coolant system **10** may further include a control valve **30** allowing the battery coolant line **11**, the power electronics coolant line **12**, and the radiator coolant line **13** to be

fluidly connected to or separated from each other. That is, the battery coolant line **11**, the power electronics coolant line **12**, and the radiator coolant line **13** may be selectively connected to or separated from each other by various switching operations of the control valve **30** so that the control valve **30** may be configured to control the flow of a coolant between the battery coolant line **11**, the power electronics coolant line **12**, and the radiator coolant line **13**.
(13) The control valve **30** may include a valve housing having a plurality of ports **31**, **32**, **33**, **34**, **35**, and **36** connected to the battery coolant line **11**, the power electronics coolant line **12**, and the radiator coolant line **13**, and a cylindrical valve member rotatable to selectively open or close the plurality of ports **31**, **32**, **33**, **34**, **35**, and **36** in the valve housing. The valve member may have a plurality of passages therein, and the rotation of the valve member may cause the passages of the valve member to selectively communicate with the plurality of ports **31**, **32**, **33**, **34**, **35**, and **36** so that the plurality of ports **31**, **32**, **33**, **34**, **35**, and **36** may be individually opened or closed, or some of the ports may communicate with each other.
(14) Specifically, the control valve **30** may include a first battery-side port **31** connected to an inlet of the battery coolant line **11**, a second battery-side port **32** connected to an outlet of the battery coolant line **11**, a first power electronic-side port **33** connected to an inlet of the power electronics coolant line **12**, a second power electronic-side port **34** connected to an outlet of the power electronics coolant line **12**, a first radiator-side port **35** connected to an inlet of the radiator coolant line **13**, and a second radiator-side port **36** connected to an outlet of the radiator coolant line **13**.
(15) The control valve **30** may be configured to perform switching operations under control of a controller **100** so that at least some of the first battery-side port **31**, the second battery-side port **32**, the first power electronic-side port **33**, the second power electronic-side port **34**, the first radiator-side port **35**, or the second radiator-side port **36** may be allowed to selectively communicate with each other. Accordingly, the battery coolant line **11**, the power electronics coolant line **12**, and the radiator coolant line **13** may be selectively connected to or separated from each other, and the coolant system **10** may operate in any one of a plurality of circulation modes forming various flows of the coolant.
(16) The coolant system **10** may further include a reservoir **28** fluidly connected to the control valve **30**, and the reservoir **28** may temporarily store and replenish the coolant so that the circulation rate of the coolant may be kept constant.
(17) The battery coolant line **11** may be fluidly connected to a battery pump **15**. The battery **21**, a battery warmer **23**, and a battery chiller **22** may be fluidly connected to the battery coolant line **11**.
(18) The battery **21** may have a coolant passage provided inside or outside thereof, and the coolant may pass through the coolant passage. The battery coolant line **11** may be fluidly connected to the coolant passage of the battery **21**.
(19) The battery warmer **23** may be disposed on the upstream or downstream side of the battery **21**, and the battery warmer **23** may heat the coolant circulating in the battery coolant line **11** so that the battery **21** may be warmed up by the heated coolant. In some examples, the battery warmer **23** may be an electric heater. In some examples, the battery warmer **23** may be a heater configured to heat a battery coolant by exchanging heat with a high-temperature fluid. Referring to FIG. **1**, the battery warmer **23** may be disposed on the downstream side of the battery **21** in a coolant flow direction.
(20) The battery pump **15** may allow the coolant to circulate, and an inlet of the battery pump **15** may communicate with the first battery-side port **31** of the control valve **30**. Accordingly, the coolant discharged from the first battery-side port **31** of the control valve **30** may be sucked into the inlet of the battery pump **15**. Referring to FIG. **1**, the battery pump **15** may be disposed on the upstream side of the battery **21**.
(21) The battery chiller **22** may be disposed on the upstream or downstream side of the battery **21**, and the battery chiller **22** may be configured to thermally connect the HVAC system **40** and the battery coolant line **11**. Referring to FIG. **1**, the battery chiller **22** may include a first passage **22*a*** fluidly connected to the battery coolant line **11**, and a second passage **22*b*** fluidly connected to a

distribution line **57** of the HVAC system **40** to be described below. The first passage **22***a* may be disposed on the downstream side of the battery warmer **23**, and the coolant passing through the first passage **22***a* may release heat to a refrigerant passing through the second passage **22***b* so that the coolant may be cooled in the battery chiller **22**, and the refrigerant may be heated or evaporated in the battery chiller **22**. An outlet of the first passage **22***a* may communicate with the second battery-side port **32** of the control valve **30**, and accordingly the coolant discharged from the first passage **22***a* may be directed to the second battery-side port **32** of the control valve **30**.

(22) The power electronics coolant line **12** may be fluidly connected to a power electronic pump **16**. The power electronic components **25***a* and **25***b* and a heat exchanger **26** may be fluidly connected to the power electronics coolant line **12**.

(23) Each of the power electronic components **25***a* and **25***b* may have a coolant passage provided inside or outside thereof, and the coolant may pass through the coolant passage. The power electronics coolant line **12** may be fluidly connected to the coolant passage of the power electronic component.

(24) The power electronic components **25***a* and **25***b* may be various components such as an autonomous driving controller, an integrated charging control unit, an inverter, and an electric motor. Referring to FIG. **1**, the power electronic components **25***a* and **25***b* may include an electric motor **25***a*, and an inverter **25***b* controlling the electric motor **25***a*.

(25) A heat sink temperature sensor **29** may be disposed on a heat sink **25***c* of any one of the power electronic components **25***a* and **25***b*. When the temperature of the heat sink **25***c* is kept constant (for example, the vehicle is traveling at a constant speed), the controller **100** may be configured to determine the temperature of the power electronics coolant line **12** based on a temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29**.

(26) In some examples, the inverter **25***b* may have the heat sink **25***c* attached to a circuit board thereof, the heat sink temperature sensor **29** may be disposed on the heat sink **25***c* of the inverter **25***b*, and the heat sink temperature sensor **29** may sense the temperature of the heat sink **25***c* of the inverter **25***b*. The power electronics coolant line **12** may be fluidly connected to the heat sink **25***c* of the inverter **25***b*.

(27) When the vehicle is running at a constant speed, the temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** may be kept constant or a variation in the temperature of the heat sink **25***c* may be less than a predetermined threshold (e.g., 2° C.). In this case, the temperature of the coolant passing through the power electronics coolant line **12** may be similar to the temperature of the heat sink **25***c* with an error of about 1° C. That is, when the temperature of the heat sink **25***c* is kept constant or the variation in the temperature of the heat sink **25***c* is less than the predetermined threshold (e.g., 2° C.), the temperature of the coolant passing through the power electronics coolant line **12** may be determined based on the temperature sensed by the heat sink temperature sensor **29**.

(28) For example, the controller **100** may determine the temperature sensed by the heat sink temperature sensor **29** as the temperature of the coolant passing through the power electronics coolant line **12**, or may correct the temperature sensed by the heat sink temperature sensor **29** and determine the corrected temperature as the temperature of the coolant passing through the power electronics coolant line **12**.

(29) The power electronic pump **16** may be configured to allow the coolant to circulate, and an inlet of the power electronic pump **16** may communicate with the first power electronic-side port **33** of the control valve **30**. Accordingly, the coolant discharged from the first power electronic-side port **33** of the control valve **30** may be sucked into the inlet of the power electronic pump **16**. Referring to FIG. **1**, the power electronic pump **16** may be disposed on the upstream side of the power electronic components **25***a* and **25***b*.

(30) The heat exchanger **26** may be disposed on the upstream or downstream side of the power electronic components **25***a* and **25***b*, and the heat exchanger **26** may be configured to thermally

connect the HVAC system **40** and the power electronics coolant line **12**.

(31) Referring to FIG. **1**, the heat exchanger **26** may include a first passage **26***a* fluidly connected to the power electronics coolant line **12**, and a second passage **26***b* fluidly connected to a third line **53** of the HVAC system **40** to be described below. The first passage **26***a* may be disposed on the downstream side of the power electronic components **25***a* and **25***b*, and the coolant passing through the first passage **26***a* may release heat to the refrigerant passing through the second passage **26***b* so that the coolant may be cooled in the heat exchanger **26**, and the refrigerant may be heated or evaporated in the heat exchanger **26**. An outlet of the first passage **26***a* may communicate with the second power electronic-side port **34** of the control valve **30**, and accordingly the coolant discharged from the first passage **26***a* may be directed to the second power electronic-side port **34** of the control valve **30**.

(32) The radiator coolant line **13** may be configured to connect the control valve **30** and the radiator **27**. The inlet of the radiator coolant line **13** may communicate with the first radiator-side port **35** of the control valve **30**, and accordingly the coolant discharged from the first radiator-side port **35** of the control valve **30** may be directed to an inlet of the radiator **27**. The outlet of the radiator coolant line **13** may communicate with the second radiator-side port **36** of the control valve **30**, and accordingly the coolant discharged from an outlet of the radiator **27** may be directed to the second radiator-side port **36** of the control valve **30**.

(33) The radiator **27** may be disposed adjacent to a front grille **17** of the vehicle, and the radiator **27** may be configured to transfer heat between the coolant passing through an internal passage thereof and air passing by an exterior surface of the radiator **27**.

(34) The controller **100** may control the operation of the control valve **30** and the operation of an active air flap **18** based on temperatures of the power electronic components **52***a* and **52***b*, a temperature of the battery **21**, an ambient temperature, operating conditions of the HVAC system **40**, and the like. The controller **100** may estimate or predict the temperatures of the power electronic components **52***a* and **52***b* based on the temperature of the coolant passing through the power electronics coolant line **12**. In particular, when the HVAC system **40** operates in a heating mode, the controller **100** may control the operation of the control valve **30** and the operation of the active air flap **18** based on the temperatures of the power electronic components **52***a* and **52***b*, which are main heat absorption sources, and the temperature of the coolant passing through the power electronics coolant line **12** so as to improve heating efficiency of the HVAC system **40**.

(35) The coolant system **10** may have a temperature sensor disposed in at least one of the battery coolant line **11**, the power electronics coolant line **12**, or the radiator coolant line **13**. The controller **100** may be configured to control the control valve **30**, the active air flap **18**, a cooling fan **19**, the battery pump **15**, and the power electronic pump **16** based on a temperature sensed by the temperature sensor.

(36) The controller **100** may control the switching operation(s) of the control valve **30** under predetermined condition(s) in a manner that fluidly connects a coolant line without any temperature sensor to a coolant line with the temperature sensor, and determine the temperature of the coolant passing through the coolant line without any temperature sensor based on the temperature sensed by the temperature sensor.

(37) In some examples, as illustrated in FIG. **1**, one temperature sensor **81** may be disposed in the battery coolant line **11**, and no temperature sensor may be provided in the power electronics coolant line **12** and the radiator coolant line **13**. In some examples, the temperature sensor **81** may be located between an outlet of the battery pump **15** and an inlet of the coolant passage of the battery **21** in the battery coolant line **11**.

(38) The controller **100** may control the switching operation of the control valve **30** under a predetermined condition in a manner that fluidly connects the power electronics coolant line **12** to the battery coolant line **11**, and determine the temperature of the coolant passing through the power electronics coolant line **12** based on a temperature sensed by the temperature sensor **81**.

(39) The controller **100** may control the switching operation of the control valve **30** under a predetermined condition in a manner that fluidly connects the radiator coolant line **13** to the battery coolant line **11**, and determine the temperature of the coolant passing through the radiator coolant line **13** based on the temperature sensed by the temperature sensor **81**.
(40) In some examples, as illustrated in FIG. **2**, a first temperature sensor **81** may be disposed in the battery coolant line **11**, and a second temperature sensor **82** may be disposed in the radiator coolant line **13**. No temperature sensor may be provided in the power electronics coolant line **12**. The controller **100** may be configured to control the control valve **30**, the active air flap **18**, the cooling fan **19**, the battery pump **15**, and the power electronic pump **16** based on a temperature of the coolant sensed by the first temperature sensor **81** or the second temperature sensor **82**. In some examples, the first temperature sensor **81** may be located between the outlet of the battery pump **15** and the inlet of the coolant passage of the battery **21**, and the second temperature sensor **82** may be located between the outlet of the radiator **27** and the second radiator-side port **36** of the control valve **30**.
(41) In some implementations, as illustrated in FIG. **2**, the controller **100** may control the switching operation(s) of the control valve **30** under predetermined condition(s) in a manner that fluidly connects the power electronics coolant line **12** to the battery coolant line **11** and/or the radiator coolant line **13**, and determine the temperature of the coolant passing through the power electronics coolant line **12** based on the temperature sensed by the first temperature sensor **81** or the second temperature sensor **82**.
(42) By various switching operations of the control valve **30**, the coolant in the coolant system **10** may be allowed to selectively circulate in the battery coolant line **11**, the power electronics coolant line **12**, and/or the radiator coolant line **13** according to various circulation modes.
(43) FIG. **3** illustrates a state in which the coolant in the coolant system **10** circulates in a first circulation mode as the control valve **30** performs a first switching operation. The first circulation mode refers to a mode in which the battery coolant line **11** is fluidly separated from the power electronics coolant line **12** and the radiator coolant line **13**, and the power electronics coolant line **12** is fluidly connected to the radiator coolant line **13**.
(44) Referring to FIG. **3**, the control valve **30** may perform the first switching operation so that the first battery-side port **31** may communicate with the second battery-side port **32** (see a**1**), the first power electronic-side port **33** may communicate with the second radiator-side port **36** (see a**2**), and the second power electronic-side port **34** may communicate with the first radiator-side port **35** (see a**3**). Accordingly, the battery coolant line **11** may be fluidly separated from the power electronics coolant line **12** and the radiator coolant line **13**, and the power electronics coolant line **12** may be fluidly connected to the radiator coolant line **13**.
(45) Referring to FIG. **3**, a portion of the coolant may circulate in the battery coolant line **11**. Specifically, a portion of the coolant may sequentially pass through the coolant passage of the battery **21**, a coolant passage of the battery warmer **23**, and the first passage **22***a* of the battery chiller **22** by the battery pump **15**. A remaining portion of the coolant may circulate in the power electronics coolant line **12** and the radiator coolant line **13**. Specifically, the remaining portion of the coolant may sequentially pass through the coolant passages of the power electronic components **25***b* and **25***a*, the first passage **26***a* of the heat exchanger **26**, and the internal passage of the radiator **27** by the power electronic pump **16**. Accordingly, a portion of the coolant may independently circulate through the battery coolant line **11**, and the remaining portion of the coolant may independently circulate through the power electronics coolant line **12** and the radiator coolant line **13**.
(46) FIG. **4** illustrates a state in which the coolant in the coolant system **10** circulates in a second circulation mode as the control valve **30** performs a second switching operation. The second circulation mode refers to a mode in which the radiator coolant line **13** is fluidly connected to the battery coolant line **11**, the battery coolant line **11** is fluidly connected to the power electronics

coolant line **12**, and the power electronics coolant line **12** is fluidly connected to the radiator coolant line **13**. That is, the second circulation mode refers to a mode in which the battery coolant line **11**, the power electronics coolant line **12**, and the radiator coolant line **13** are fluidly connected to each other.

(47) Referring to FIG. **4**, the control valve **30** may perform the second switching operation so that the first battery-side port **31** may communicate with the second radiator-side port **36** (see b**1**), the second battery-side port **32** may communicate with the first power electronic-side port **33** (see b**2**), and the second power electronic-side port **34** may communicate with the first radiator-side port **35** (see b**3**). Accordingly, the radiator coolant line **13** may be fluidly connected to the battery coolant line **11**, the battery coolant line **11** may be fluidly connected to the power electronics coolant line **12**, and the power electronics coolant line **12** may be fluidly connected to the radiator coolant line **13**.

(48) Referring to FIG. **4**, the coolant may sequentially pass through the coolant passage of the battery **21**, the coolant passage of the battery warmer **23**, and the first passage **22***a* of the battery chiller **22** by the battery pump **15**, and the coolant discharged from the first passage **22***a* of the battery chiller **22** may sequentially pass through the coolant passages of the power electronic components **25***b* and **25***a*, the first passage **26***a* of the heat exchanger **26**, and the internal passage of the radiator **27** by the power electronic pump **16**.

(49) FIG. **5** illustrates a state in which the coolant in the coolant system **10** circulates in a third circulation mode as the control valve **30** performs a third switching operation. The third circulation mode refers to a mode in which the radiator coolant line **13** is fluidly blocked from the control valve **30**, and the battery coolant line **11** is fluidly separated from the power electronics coolant line **12**.

(50) Referring to FIG. **5**, the control valve **30** may perform the third switching operation so that the first radiator-side port **35** and the second radiator-side port **36** may be closed, the first battery-side port **31** may communicate with the second battery-side port **32** (see c**1**), and the first power electronic-side port **33** may communicate with the second power electronic-side port **34** (see c**2**). Accordingly, the radiator coolant line **13** may be fluidly blocked from the control valve **30**, and the battery coolant line **11** may be fluidly separated from the power electronics coolant line **12**.

(51) Referring to FIG. **5**, a portion of the coolant may sequentially pass through the coolant passage of the battery **21**, the coolant passage of the battery warmer **23**, and the first passage **22***a* of the battery chiller **22** by the battery pump **15**. A remaining portion of the coolant may sequentially pass through the coolant passages of the power electronic components **25***b* and **25***a* and the first passage **26***a* of the heat exchanger **26** by the power electronic pump **16**. Accordingly, a portion of the coolant may independently circulate through the battery coolant line **11**, and the remaining portion of the coolant may independently circulate through the power electronics coolant line **12**.

(52) FIG. **6** illustrates a state in which the coolant in the coolant system **10** circulates in a fourth circulation mode as the control valve **30** performs a fourth switching operation. The fourth circulation mode refers to a mode in which the radiator coolant line **13** is fluidly blocked from the control valve **30**, and the battery coolant line **11** is fluidly connected to the power electronics coolant line **12**.

(53) Referring to FIG. **6**, the control valve **30** may perform the fourth switching operation so that the first radiator-side port **35** and the second radiator-side port **36** may be closed, the first battery-side port **31** may communicate with the second power electronic-side port **34** (see d**1**), and the second battery-side port **32** may communicate with the first power electronic-side port **33** (see d**2**). Accordingly, the radiator coolant line **13** may be fluidly blocked from the control valve **30**, and the battery coolant line **11** may be fluidly connected to the power electronics coolant line **12**.

(54) Referring to FIG. **6**, the coolant may sequentially pass through the coolant passage of the battery **21**, the coolant passage of the battery warmer **23**, and the first passage **22***a* of the battery chiller **22** by the battery pump **15**, and the coolant discharged from the first passage **22***a* of the

battery chiller **22** may sequentially pass through the coolant passages of the power electronic components **25***b* and **25***a* and the first passage **26***a* of the heat exchanger **26** by the power electronic pump **16**.
(55) The HVAC system **40** may be thermally connected to a passenger compartment. In particular, the HVAC system **40** may be configured to heat or cool the air in the passenger compartment of the vehicle using a refrigerant.
(56) A refrigerant circulation path **50** may be configured to provide various circulation paths based on various operating modes such as an operating mode of the HVAC system **40**, cooling and warming-up of the battery **21**, and cooling and warming-up of the power electronic components **25***a* and **25***b*.
(57) The HVAC system **40** may include a compressor **41**, an interior condenser **42**, a heating-side expansion valve **43**, an exterior heat exchanger **44**, a cooling-side expansion valve **45**, and an evaporator **46** fluidly connected through the refrigerant circulation path **50**.
(58) The compressor **41** may be configured to compress the refrigerant and allow the refrigerant to circulate. In some examples, the compressor **41** may be an electric compressor driven by electric energy.
(59) The interior condenser **42** may be disposed on the downstream side of the compressor **41**, and the interior condenser **42** may be configured to condense the refrigerant received from the compressor **41**. That is, the refrigerant compressed by the compressor **41** may transfer heat to the air and be condensed in the interior condenser **42**. Accordingly, the interior condenser **42** may heat the air using the refrigerant compressed by the compressor **41**, and the air heated by the interior condenser **42** may be directed into the passenger compartment.
(60) The heating-side expansion valve **43** may be disposed on the downstream side of the interior condenser **42** in the refrigerant circulation path **50**. Specifically, the heating-side expansion valve **43** may be disposed between the interior condenser **42** and the exterior heat exchanger **44**. During a heating operation of the HVAC system **40**, the heating-side expansion valve **43** may adjust the flow of the refrigerant and/or the flow rate of the refrigerant into the exterior heat exchanger **44**. The heating-side expansion valve **43** may be configured to expand the refrigerant received from the interior condenser **42** during the heating operation of the HVAC system **40**. The opening degree of the heating-side expansion valve **43** may be varied by the controller **100**. As the opening degree of the heating-side expansion valve **43** is varied, the flow rate of the refrigerant into the exterior heat exchanger **44** may be varied. That is, the heating-side expansion valve **43** may be controlled by the controller **100** during the heating operation of the HVAC system **40**.
(61) In some examples, the heating-side expansion valve **43** may be an electronic expansion valve (EXV) having an actuator **43***a*. The actuator **43***a* may have a shaft which is movable to open or close an orifice defined in a valve body of the heating-side expansion valve **43**, and the position of the shaft may be varied depending on the rotation direction, rotation degree, and the like of the actuator **43***a*, and accordingly the opening degree of the orifice of the heating-side expansion valve **43** may be varied. The controller **100** may control the operation of the actuator **43***a*. The heating-side expansion valve **43** may be a full open type EXV. When the HVAC system **40** does not operate in the heating mode, the heating-side expansion valve **43** may be fully opened (the opening degree of the heating-side expansion valve **43** is 100%) so that the refrigerant may pass through the heating-side expansion valve **43**, and thus the refrigerant may be prevented from being expanded by the heating-side expansion valve **43**.
(62) The exterior heat exchanger **44** together with the radiator **27** may be disposed adjacent to the front grille **17** of the vehicle. The exterior heat exchanger **44** may be configured to transfer heat between the coolant passing through an internal passage thereof and the air passing by an exterior surface of the exterior heat exchanger **44**.
(63) In some examples, the cooling fan **19** may be located behind the exterior heat exchanger **44** and the radiator **27**, and the exterior heat exchanger **44** and the radiator **27** may exchange heat with

the ambient air forcibly blown by the cooling fan **19** so that a heat transfer rate of the exterior heat exchanger **44** and the radiator **27** may be further increased.

(64) In some examples, the active air flap **18** may be configured to adjust the opening degree of the front grille **17**, and the active air flap **18** may be disposed between the front grille **17** and the exterior heat exchanger **44**. Each flap of the active air flap **18** may be rotatably mounted to adjust the opening degree of a corresponding opening of the front grille **17**. The controller **100** may monitor the vehicle speed and the opening degree of the front grille **17**, receive the vehicle speed from a vehicle controller, and adjust the opening degree of the front grille **17** by the active air flap **18**. The vehicle speed and the opening degree of the front grille **17** may be used to calculate the flow rate of air passing by the exterior surface of the exterior heat exchanger **44** and the exterior surface of the radiator **27**.

(65) During a cooling operation of the HVAC system **40**, the exterior heat exchanger **44** may be configured to condense the refrigerant received from the interior condenser **42**. That is, the refrigerant passing through the exterior heat exchanger **44** may serve as an exterior condenser that condenses the refrigerant by transferring heat to the ambient air during the cooling operation of the HVAC system **40**. During the heating operation of the HVAC system **40**, the exterior heat exchanger **44** may be configured to evaporate the refrigerant expanded by the heating-side expansion valve **43**. That is, the refrigerant passing through the exterior heat exchanger **44** may serve as an exterior evaporator that evaporates the refrigerant by absorbing heat from the ambient air during the heating operation of the HVAC system **40**. In particular, the exterior heat exchanger **44** may exchange heat with the ambient air forcibly blown by the cooling fan **19** so that a heat transfer rate between the exterior heat exchanger **44** and the ambient air may be further increased.

(66) The cooling-side expansion valve **45** may be disposed on the downstream side of the exterior heat exchanger **44** in the refrigerant circulation path **50**, and the cooling-side expansion valve **45** may be disposed between the exterior heat exchanger **44** and the evaporator **46** in the refrigerant circulation path **50**. The cooling-side expansion valve **45** may be disposed on the upstream side of the evaporator **46** so that it may adjust the flow of the refrigerant and/or the flow rate of the refrigerant into the evaporator **46**. During the cooling operation of the HVAC system **40**, the cooling-side expansion valve **45** may be configured to expand the refrigerant received from the exterior heat exchanger **44**.

(67) In some examples, the cooling-side expansion valve **45** may be a thermal expansion valve (TXV) which senses the temperature and/or pressure of the refrigerant and adjusts the opening degree of the cooling-side expansion valve **45**. Specifically, the cooling-side expansion valve **45** may be a TXV having a solenoid valve **45***a* selectively blocking or unblocking the flow of the refrigerant into an internal passage of the cooling-side expansion valve **45**. The solenoid valve **45***a* may be opened or closed by the controller **100**, thereby unblocking or blocking the flow of the refrigerant into the cooling-side expansion valve **45**. When the solenoid valve **45***a* is opened, the refrigerant may be allowed to flow into the cooling-side expansion valve **45**, and when the solenoid valve **45***a* is closed, the refrigerant may be blocked from flowing into the cooling-side expansion valve **45**. In some examples, the solenoid valve **45***a* may be integrally formed in a valve body of the cooling-side expansion valve **45** to thereby open or close the internal passage of the cooling-side expansion valve **45**. In some examples, the solenoid valve **45***a* may be disposed on the upstream side of the cooling-side expansion valve **45** to thereby selectively open or close an inlet of the cooling-side expansion valve **45**. When the solenoid valve **45***a* is closed, the refrigerant may be blocked from flowing into the cooling-side expansion valve **45** and the evaporator **46**, and accordingly the cooling operation of the HVAC system **40** may not be performed. When the solenoid valve **45***a* is opened, the refrigerant may be directed to the cooling-side expansion valve **45** and the evaporator **46**. That is, when the solenoid valve **45***a* of the cooling-side expansion valve **45** is opened to a predetermined degree, the cooling operation of the HVAC system **40** may be performed.

(68) The HVAC system **40** may include an HVAC case **70** including a blower blowing the heated air or cooled air into the passenger compartment. The HVAC case **70** may have an inlet and an outlet, and the HVAC case **70** may be configured to allow the air to be directed into the passenger compartment of the vehicle. The evaporator **46** and the interior condenser **42** may be located in the HVAC case **70**. An air mixing door **71** may be disposed between the evaporator **46** and the interior condenser **42**, and an electric heater **72** such as a PTC (positive temperature coefficient) heater may be disposed on the downstream side of the interior condenser **42** in an air flow direction.
(69) The evaporator **46** may be configured to evaporate the refrigerant, thereby cooling the air directed toward the passenger compartment, and the interior condenser **42** may be configured to condense the refrigerant, thereby heating the air directed toward the passenger compartment. The electric heater **72** may be configured to heat the air directed toward the passenger compartment using electric energy.
(70) The air mixing door **71** may be disposed between the evaporator **46** and the interior condenser **42**. As the position of the air mixing door **71** changes, the flow rate of the air cooled by the evaporator **46** and the flow rate of the air heated by the interior condenser **42** may be mixed at a predetermined ratio. The air cooled by the evaporator **46**, the air heated by the interior condenser **42**, and the air mixed by the air mixing door **71** in the HVAC case **70** may be directed toward a front seat area of the passenger compartment through an air distributer unit.
(71) The HVAC system **40** may further include an accumulator **47** located on the upstream side of the compressor **41**. The accumulator **47** may separate a liquid refrigerant from the refrigerant, thereby preventing the liquid refrigerant from flowing into the compressor **41**.
(72) The refrigerant circulation path **50** may include a first line **51** extending from an outlet of the compressor **41** to the interior condenser **42**, a second line **52** extending from the interior condenser **42** to the heating-side expansion valve **43**, a third line **53** extending from the heating-side expansion valve **43** to the exterior heat exchanger **44**, a fourth line **54** extending from the exterior heat exchanger **44** to the cooling-side expansion valve **45**, a fifth line **55** extending from the cooling-side expansion valve **45** to the evaporator **46**, and a sixth line **56** extending from the evaporator **46** to the compressor **41**.
(73) The HVAC system **40** may further include the distribution line **57** configured to allow at least a portion of the refrigerant discharged from the exterior heat exchanger **44** to be directed from the upstream side of the cooling-side expansion valve **45** to the compressor **41**. The distribution line **57** may be configured to connect an upstream point **54***a* of the cooling-side expansion valve **45** and an upstream point of the compressor **41**. An inlet of the distribution line **57** may be connected to the fourth line **54** of the refrigerant circulation path **50** at the upstream point **54***a* of the cooling-side expansion valve **45**. An outlet of the distribution line **57** may be connected to the refrigerant circulation path **50** at the upstream point of the compressor **41**. Specifically, the outlet of the distribution line **57** may be connected to the sixth line **56** of the refrigerant circulation path **50** at an upstream point **56***a* of the accumulator **47** located on the upstream side of the compressor **41**. Accordingly, at least a portion of the refrigerant may be directed to the compressor **41** through the distribution line **57**.
(74) The distribution line **57** may be configured to allow at least a portion of the refrigerant discharged from the exterior heat exchanger **44** to be directed toward the compressor **41** by bypassing the cooling-side expansion valve **45** and the evaporator **46**. Accordingly, the refrigerant may be distributed to the distribution line **57** and the fifth line **55** at a predetermined ratio. The second passage **22***b* of the battery chiller **22** may be fluidly connected to the distribution line **57**, and the refrigerant passing through the second passage **22***b* of the battery chiller **22** may absorb heat from the coolant passing through the first passage **22***a* of the battery chiller **22** (heat absorption) so that the refrigerant may be heated or evaporated in the battery chiller **22**, and the coolant may be cooled in the battery chiller **22**.
(75) The HVAC system **40** may further include a bypass line **61** configured to allow at least a

portion of the refrigerant passing through the distribution line **57** to be directed from the upstream side of the second passage **22***b* of the battery chiller **22** to the downstream side of the second passage **22***b* of the battery chiller **22**. The bypass line **61** may be configured to connect an upstream point of the second passage **22***b* of the battery chiller **22** and an upstream point **56***b* of the compressor **41**. An inlet of the bypass line **61** may be connected to the distribution line **57** at the upstream point of the second passage **22***b* of the battery chiller **22**, and an outlet of the bypass line **61** may be connected to the sixth line **56** of the refrigerant circulation path **50** at the upstream point **56***b* of the compressor **41**. Accordingly, a portion of the refrigerant passing through the distribution line **57** may bypass the second passage **22***b* of the battery chiller **22** through the bypass line **61** so that it may be directed to the compressor **41**.
(76) The HVAC system **40** may further include a battery-side expansion valve **49** located on the upstream side of the battery chiller **22** in the distribution line **57**.
(77) The battery-side expansion valve **49** may be configured to adjust the flow of the refrigerant and/or the flow rate of the refrigerant into the second passage **22***b* of the battery chiller **22**, and to expand the refrigerant received from the exterior heat exchanger **44**. In some examples, the battery-side expansion valve **49** may be an EXV having an actuator **49***d*. The controller **100** may control the operation of the actuator **49***d*.
(78) The battery-side expansion valve **49** may include an inlet port **49***a* communicating with the exterior heat exchanger **44**, and an outlet port **49***b* communicating with the second passage **22***b* of the battery chiller **22**. The battery-side expansion valve **49** may include a valve housing, and a valve member moved by the actuator **49***d* in the valve housing.
(79) The valve member may be configured to open or close the inlet port **49***a* by the actuator **49***d*. When the inlet port **49***a* is opened, the inlet port **49***a* may receive the refrigerant discharged from the exterior heat exchanger **44**.
(80) The valve member may be configured to adjust the opening degree of the outlet port **49***b* by the actuator **49***d*. When the temperature of the battery exceeds a threshold temperature (that is, waste heat of the battery is relatively high, so the cooling of the battery is provided), the opening degree of the outlet port **49***b* may be adjusted by the valve member and the actuator **49***d* to correspond to a cooling load of the battery **21** so that the refrigerant may be expanded at the outlet port **49***b*, and the flow rate of the refrigerant into the second passage **22***b* of the battery chiller **22** may be adjusted, and accordingly the expanded refrigerant may be directed to the second passage **22***b* of the battery chiller **22**. The refrigerant passing through the second passage **22***b* of the battery chiller **22** may directly absorb heat from the coolant passing through the first passage **22***a* of the battery chiller **22** so that the refrigerant may be evaporated in the second passage **22***b* of the battery chiller **22**. When the cooling of the battery **21** is provided, the opening degree of the outlet port **49***b* may be adjusted so that the outlet port **49***b* may serve as an expansion valve that expands the refrigerant flowing into the second passage **22***b* of the battery chiller **22**.
(81) In some examples of the present disclosure, the battery-side expansion valve **49** may further include a bypass port **49***c* communicating with the inlet of the bypass line **61**, and the valve member may be configured to adjust the opening degree of the bypass port **49***c* by the actuator **49***d*. When the bypass port **49***c* is opened, the refrigerant discharged from the bypass port **49***c* may be directed to the compressor **41** through the bypass line **61**.
(82) The bypass port **49***c* may be directly connected to the bypass line **61**, and the opening degree of the bypass port **49***c* may be adjusted by the actuator **49***d* so that the flow rate of the refrigerant into the bypass line **61** may be adjusted. The bypass port **49***c* may serve as a flow control valve that adjusts the flow rate of the refrigerant bypassing the second passage **22***b* of the battery chiller **22**.
(83) The opening degree of the outlet port **49***b* and the opening degree of the bypass port **49***c* may be adjusted depending on the temperature of the battery, and accordingly the flow rate of the refrigerant into the second passage **22***b* of the battery chiller **22** and the flow rate of the refrigerant passing through the bypass line **61** may be adjusted at a predetermined ratio. For example, when

the temperature of the battery is lower than or equal to a threshold temperature (that is, the waste heat of the battery is relatively reduced), the opening degree of the outlet port **49***b* may be relatively reduced, and the opening degree of the bypass port **49***c* may be relatively increased so that the flow rate of the refrigerant into the second passage **22***b* of the battery chiller **22** may be lower than the flow rate of the refrigerant passing through the bypass line **61**.
(84) The second passage **26***b* of the heat exchanger **26** may be fluidly connected to the third line **53** of the refrigerant circulation path **50** of the HVAC system **40**. Accordingly, at least a portion of the refrigerant may pass through the second passage **26***b* of the heat exchanger **26** through the third line **53** of the refrigerant circulation path **50**. That is, the second passage **26***b* may be fluidly connected to the refrigerant circulation path **50** between the heating-side expansion valve **43** and the exterior heat exchanger **44**.
(85) When the HVAC system **40** performs the heating operation, the refrigerant may be expanded by the heating-side expansion valve **43** and then pass through the second passage **26***b* of the heat exchanger **26**, and the refrigerant passing through the second passage **26***b* may absorb heat from the coolant passing through the first passage **26***a* of the heat exchanger **26** so that the refrigerant may be evaporated in the heat exchanger **26**, and the coolant may be cooled in the heat exchanger **26**. That is, the heat exchanger **26** may serve as an evaporator that evaporates the refrigerant during the heating operation of the HVAC system **40**.
(86) When the HVAC system **40** performs the cooling operation, the heating-side expansion valve **43** may be fully opened so that the refrigerant may not be expanded in the heating-side expansion valve **43**, and the refrigerant passing through the second passage **26***b* of the heat exchanger **26** may release heat to the coolant passing through the first passage **26***a* of the heat exchanger **26** (heat release) so that the refrigerant may be condensed in the heat exchanger **26**, and the coolant may be heated in the heat exchanger **26**. That is, the heat exchanger **26** may serve as a condenser that condenses the refrigerant during the cooling operation of the HVAC system **40**.
(87) The HVAC system **40** may further include a dehumidification bypass line **59** configured to allow the refrigerant discharged from the heating-side expansion valve **43** to be directed from the upstream side of the second passage **26***b* of the heat exchanger **26** to the evaporator **46**. The dehumidification bypass line **59** may be configured to connect a downstream point **53***a* of the heating-side expansion valve **43** and an upstream point **55***a* of the evaporator **46**. Specifically, the branch line **58** may be branched from a point between the heating-side expansion valve **43** and the second passage **26***b* of the heat exchanger **26**, an inlet of the dehumidification bypass line **59** may be connected to the branch line **58**, and an outlet of the dehumidification bypass line **59** may be connected to the fifth line **55** of the refrigerant circulation path **50** at the upstream point **55***a* of the evaporator **46**.
(88) The HVAC system **40** may further include a heat exchanger bypass line **60** configured to allow at least a portion of the refrigerant discharged from the heating-side expansion valve **43** to bypass the second passage **26***b* of the heat exchanger **26** and be directed to the downstream side of the second passage **26***b* of the heat exchanger **26**. The heat exchanger bypass line **60** may be configured to connect an upstream point of the second passage **26***b* of the heat exchanger **26** and a downstream point **53***b* of the second passage **26***b* of the heat exchanger **26**. Specifically, an inlet of the heat exchanger bypass line **60** may be connected to the branch line **58**, and an outlet of the heat exchanger bypass line **60** may be connected to the third line **53** of the refrigerant circulation path **50** at the downstream point **53***b* of the second passage **26***b* of the heat exchanger **26**. Accordingly, the refrigerant discharged from the heating-side expansion valve **43** may be directed from the upstream point of the second passage **26***b* of the heat exchanger **26** to the downstream point of the second passage **26***b* of the heat exchanger **26** through the branch line **58** and the heat exchanger bypass line **60** so that at least a portion of the refrigerant discharged from the heating-side expansion valve **43** may bypass the second passage **26***b* of the heat exchanger **26**.
(89) The HVAC system **40** may further include a first control valve **63** disposed at a point to which

the branch line **58**, the dehumidification bypass line **59**, and the heat exchanger bypass line **60** are connected. The first control valve **63** may be configured to control the flow of the refrigerant (the flow direction of the refrigerant, the flow rate of the refrigerant, and the like) between the heating-side expansion valve **43**, the evaporator **46**, and the exterior heat exchanger **44**. The first control valve **63** may be configured to control the flow of the refrigerant in a manner that allows the refrigerant discharged from the heating-side expansion valve **43** to be selectively directed toward at least one of the evaporator **46**, the second passage **26***b* of the heat exchanger **26**, or the downstream point **53***b* of the second passage **26***b* of the heat exchanger **26**.

(90) The first control valve **63** may include an inlet port **63***a* communicating with the branch line **58**, a first outlet port **63***b* communicating with the heat exchanger bypass line **60**, and a second outlet port **63***c* communicating with the dehumidification bypass line **59**.

(91) In a state in which the first control valve **63** performs a first switching operation to allow the inlet port **63***a* to be discharged from the heating-side closed, the refrigerant expansion valve **43** may be directed to the second passage **26***b* of the heat exchanger **26**.

(92) In a state in which the first control valve **63** performs a second switching operation to allow the inlet port **63***a* to communicate with the first outlet port **63***b*, the refrigerant discharged from the heating-side expansion valve **43** may be directed to the downstream point **53***b* of the second passage **26***b* of the heat exchanger **26** through the heat exchanger bypass line **60**. In addition, the refrigerant may be directed from the downstream point **53***b* of the second passage **26***b* of the heat exchanger **26** to the exterior heat exchanger **44**.

(93) In a state in which the first control valve **63** performs a third switching operation to allow the inlet port **63***a* to communicate with the second outlet port **63***c*, the refrigerant discharged from the heating-side expansion valve **43** may be directed to the evaporator **46** through the dehumidification bypass line **59**. When dehumidification is provided in the passenger compartment during the heating operation of the HVAC system **40**, the first control valve **63** may perform the third switching operation so that at least a portion of the refrigerant discharged from the heating-side expansion valve **43** may be directed to the evaporator **46** through the dehumidification bypass line **59**. Accordingly, the refrigerant directed to the evaporator **46** may cool the air passing by an exterior surface of the evaporator **46** so that the dehumidification of the air flowing into the passenger compartment may be performed.

(94) The HVAC system **40** may further include a heating bypass line **62** configured to allow the refrigerant discharged from the second passage **26***b* of the heat exchanger **26** to be directed to the compressor **41**. The heating bypass line **62** may be configured to connect a downstream point of the second passage **26***b* of the heat exchanger **26** and the upstream point **56***b* of the compressor **41**. Specifically, an inlet of the heating bypass line **62** may be connected to the third line **53** of the refrigerant circulation path **50** at the downstream point of the second passage **26***b* of the heat exchanger **26**, and an outlet of the heating bypass line **62** may be connected to the sixth line **56** of the refrigerant circulation path **50** at the upstream point **56***b* of the compressor **41**. Accordingly, the refrigerant discharged from the second passage **26***b* of the heat exchanger **26** may bypass the exterior heat exchanger **44**, the battery-side expansion valve **49**, and the second passage **22***b* of the battery chiller **22** through the heating bypass line **62** so that the refrigerant may be directed from the second passage **26***b* of the heat exchanger **26** to the compressor **41**.

(95) The HVAC system **40** may further include a second control valve **64** disposed at a point to which the heating bypass line **62** and the third line **53** of the refrigerant circulation path **50** are connected. The second control valve **64** may be configured to control the flow of the refrigerant (the flow direction of the refrigerant, the flow rate of the refrigerant, and the like) between the second passage **26***b* of the heat exchanger **26**, the exterior heat exchanger **44**, and the compressor **41**. The second control valve **64** may be configured to control the flow of the refrigerant in a manner that allows the refrigerant discharged from the second passage **26***b* of the heat exchanger **26** to be selectively directed toward the exterior heat exchanger **44** and/or the compressor **41**.

(96) The second control valve **64** may include an inlet port **64***a* communicating with the second passage **26***b* of the heat exchanger **26**, a first outlet port **64***b* communicating with the exterior heat exchanger **44**, and a second outlet port **64***c* communicating with the heating bypass line **62**.
(97) In a state in which the second control valve **64** performs a first switching operation to allow the inlet port **64***a* to communicate with the first outlet port **64***b*, the refrigerant discharged from the second passage **26***b* of the heat exchanger **26** may be directed to the exterior heat exchanger **44**.
(98) In a state in which the second control valve **64** performs a second switching operation to allow the inlet port **64***a* to communicate with the second outlet port **64***c*, the refrigerant discharged from the second passage **26***b* of the heat exchanger **26** may be directed to the compressor **41** through the heating bypass line **62**.
(99) The controller **100** may be configured to control respective operations of the compressor **41**, the actuator **43***a* of the heating-side expansion valve **43**, the solenoid valve **45***a* of the cooling-side expansion valve **45**, the actuator **49***d* of the battery-side expansion valve **49**, the air mixing door **71**, an actuator of the first control valve **63**, an actuator of the second control valve **64**, the battery warmer **23**, an actuator of the cooling fan **19**, the electric heater **72**, an actuator of the air mixing door **71**, an actuator of the control valve **30**, an actuator of the battery pump **15**, and an actuator of the power electronic pump **16**. Thus, the overall operation of the coolant system **10** and the HVAC system **40** may be controlled by the controller **100**. In some examples, the controller **100** may be a full automatic temperature control (FATC) system.
(100) The controller **100** may include a processor and a memory. The processor may be programmed to receive instructions stored in the memory, and transmit instructions to various actuators. The memory may be a data store such as a hard disk drive, a solid state drive, a server, a volatile storage medium, and a non-volatile storage medium.
(101) A battery management system **110** may transmit instructions for the cooling of the battery **21** and the warming-up of the battery **21** to the controller **100**, and accordingly the controller **100** may control the operation of the compressor **41** and the operation of the actuator **49***d* of battery-side expansion valve **49**.
(102) The coolant system **10** in FIG. **1** may include one temperature sensor **81** disposed in the battery coolant line **11**. When the coolant system **10** operates in the first circulation mode or the third circulation mode, the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11** and/or the radiator coolant line **13**. In this state, when the temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** is kept constant or a variation in the temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** is less than a predetermined threshold (e.g., 2° C.) (that is, the vehicle is traveling at a constant speed), the controller **100** may determine the temperature of the coolant passing through the power electronics coolant line **12** based on the temperature of the heat sink sensed by the heat sink temperature sensor **29**.
(103) While the vehicle is accelerating or decelerating, the vehicle speed may rapidly change or the torque of the electric motor **25***a* may rapidly change, and accordingly a variation in the temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** may be greater than or equal to a predetermined threshold (e.g., 2° C.) for a predetermined period of time (e.g., five seconds). In this case, the temperature of the coolant passing through the power electronics coolant line **12** may considerably differ from the temperature of the heat sink **25***c*. When the variation in the temperature of the heat sink **25***c* is greater than or equal to the predetermined threshold (e.g., 2° C.) for a predetermined period of time (e.g., five seconds), the temperature of the coolant passing through the power electronics coolant line **12** may not be determined based on the temperature sensed by the heat sink temperature sensor **29**.
(104) In a state in which the coolant system **10** including one temperature sensor **81** disposed in the battery coolant line **11** operates in the first circulation mode (see FIG. **3**), the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11**, and the power electronics

coolant line **12** may be fluidly connected to the radiator coolant line **13**. When a variation in the temperature of the heat sink **25***c* is greater than or equal to a predetermined threshold (e.g., 2° C.) for a predetermined period of time (e.g., five seconds), the control valve **30** may perform the second switching operation so that the coolant system **10** may switch from the first circulation mode to the second circulation mode (see FIG. **4**) for a predetermined period of time (e.g., ten seconds), and each of the battery pump **15** and the power electronic pump **16** may operate at maximum RPM. Accordingly, the coolant passing through the power electronics coolant line **12** may pass through the battery coolant line **11**. As a result, the temperature of the coolant passing through the power electronics coolant line **12** may become similar to a temperature sensed by the temperature sensor **81** disposed in the battery coolant line **11**, and the controller **100** may determine the temperature of the coolant passing through the power electronics coolant line **12** based on the temperature sensed by the temperature sensor **81** disposed in the battery coolant line **11**.

(105) In a state in which the coolant system **10** including one temperature sensor **81** disposed in the battery coolant line **11** operates in the third circulation mode (see FIG. **5**), the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11**, and the radiator coolant line **13** may be fluidly blocked from the control valve **30**. When a variation in the temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** is greater than or equal to a predetermined threshold (e.g., 2° C.) for a predetermined period of time (e.g., five seconds), the control valve **30** may perform the fourth switching operation so that the coolant system **10** may switch from the third circulation mode to the fourth circulation mode (see FIG. **6**) for a predetermined period of time (e.g., ten seconds), and each of the battery pump **15** and the power electronic pump **16** may operate at maximum RPM. Accordingly, the coolant passing through the power electronics coolant line **12** may pass through the battery coolant line **11**. As a result, the temperature of the coolant passing through the power electronics coolant line **12** may become similar to a temperature sensed by the temperature sensor **81** disposed in the battery coolant line **11**, and the controller **100** may determine the temperature of the coolant passing through the power electronics coolant line **12** based on the temperature sensed by the temperature sensor **81** disposed in the battery coolant line **11**.

(106) FIG. **7** illustrates a method of controlling the coolant system **10** in FIG. **1**.

(107) The controller **100** may determine whether the HVAC system **40** operates in the heating mode (S**1**).

(108) When it is determined that the HVAC system **40** operates in the heating mode, the controller **100** may determine whether the coolant system **10** operates in the first circulation mode or the third circulation mode (S**2**).

(109) In the first circulation mode, the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11**, and the power electronics coolant line **12** may be fluidly connected to the radiator coolant line **13**. In the third circulation mode, the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11**, and the radiator coolant line **13** may be fluidly blocked from the control valve **30**. That is, when the coolant system **10** operates in the first circulation mode or the third circulation mode, the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11**.

(110) When it is determined that the coolant system **10** operates in the first circulation mode or the third circulation mode, the controller **100** may determine whether a variation ΔT.sub.h in temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** is greater than or equal to a predetermined threshold A (e.g., 2° C.) for a predetermined period of time (e.g., five seconds) (S**3**). While the vehicle is running at a constant speed, the temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** may be kept constant or the variation in the temperature of the heat sink **25***c* may be less than the predetermined threshold. While the vehicle is accelerating or decelerating, the vehicle speed may rapidly change or the torque of the electric motor **25***a* may rapidly change, and accordingly the variation in the temperature of the heat sink

25*c* sensed by the heat sink temperature sensor **29** may be greater than or equal to the predetermined threshold for a predetermined period of time.
(111) When it is determined that the variation ΔT.sub.h in the temperature of the heat sink **25***c* is greater than or equal to the predetermined threshold A (e.g., 2° C.) for a predetermined period of time (e.g., five seconds), the controller **100** may control the switching operation of the control valve **30** so that the coolant system **10** may switch to the second circulation mode or the fourth circulation mode for a predetermined period of time (e.g., ten seconds) (S**4**). Here, the controller **100** may control the battery pump **15** and the power electronic pump **16** in a manner that operates the battery pump **15** and the power electronic pump **16** at maximum RPM. As the battery pump **15** and the power electronic pump **16** operate at maximum RPM, thermal equilibrium may be quickly achieved between the temperature of the coolant passing through the battery coolant line **11** and the temperature of the coolant passing through the power electronics coolant line **12**.
(112) In some examples, when it is determined that the variation ΔT.sub.h in the temperature of the heat sink **25***c* is greater than or equal to the predetermined threshold A for a predetermined period of time in a state in which the coolant system **10** operates in the first circulation mode, the controller **100** may control the control valve **30** in a manner that switches the operating mode of the coolant system **10** from the first circulation mode to the second circulation mode.
(113) In some examples, when it is determined that the variation ΔT.sub.h in the temperature of the heat sink **25***c* is greater than or equal to the predetermined threshold A for a predetermined period of time in a state in which the coolant system **10** operates in the third circulation mode, the controller **100** may control the control valve **30** in a manner that switches the operating mode of the coolant system **10** from the third circulation mode to the fourth circulation mode.
(114) In a state in which the coolant system **10** switches to the second circulation mode or the fourth circulation mode for a predetermined period of time, the controller **100** may determine a temperature T of the coolant passing through the power electronics coolant line **12** based on a temperature sensed by the temperature sensor **81** (S**5**).
(115) When it is determined that the variation ΔT.sub.h in the temperature of the heat sink **25***c* is less than the predetermined threshold A (e.g., 2° C.) for a predetermined period of time (e.g., five seconds), the controller **100** may determine the temperature T of the coolant passing through the power electronics coolant line **12** based on a temperature of the heat sink sensed by the heat sink temperature sensor **29** (S**6**).
(116) The coolant system **10** in FIG. **2** may include the first temperature sensor **81** disposed in the battery coolant line **11** and the second temperature sensor **82** disposed in the radiator coolant line **13**. When the coolant system **10** operates in the third circulation mode, the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11** and the radiator coolant line **13**. In this state, when the temperature of the heat sink sensed by the heat sink temperature sensor **29** is kept constant or a variation in the temperature of the heat sink sensed by the heat sink temperature sensor **29** is less than a predetermined threshold (e.g., 2° C.) (that is, the vehicle is running at a constant speed), the controller **100** may determine the temperature of the coolant passing through the power electronics coolant line **12** based on the temperature of the heat sink sensed by the heat sink temperature sensor **29**.
(117) While the vehicle is accelerating or decelerating, the vehicle speed may rapidly change or the torque of the electric motor **25***a* may rapidly change, and accordingly a variation in the temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** may be greater than or equal to a predetermined threshold (e.g., 2° C.) for a predetermined period of time (e.g., five seconds). In this case, the temperature of the coolant passing through the power electronics coolant line **12** may considerably differ from the temperature of the heat sink **25***c*. When the variation in the temperature of the heat sink **25***c* is greater than or equal to the predetermined threshold (e.g., 2° C.) for a predetermined period of time (e.g., five seconds), the temperature of the coolant passing through the power electronics coolant line **12** may not be determined based on the temperature

sensed by the heat sink temperature sensor **29**.

(118) In a state in which the coolant system **10** including the first temperature sensor **81** disposed in the battery coolant line **11** and the second temperature sensor **82** disposed in the radiator coolant line **13** operates in the third circulation mode (see FIG. **5**), the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11**, and the radiator coolant line **13** may be fluidly blocked from the control valve **30**. When the variation in the temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** is greater than or equal to the predetermined threshold (e.g., 2° C.) for a predetermined period of time (e.g., five seconds), the control valve **30** may perform the first switching operation or the fourth switching operation so that the coolant system **10** may switch from the third circulation mode to the first circulation mode (see FIG. **3**) or the fourth circulation mode (see FIG. **6**) for a predetermined period of time (e.g., ten seconds), and each of the battery pump **15** and the power electronic pump **16** may operate at maximum RPM. Accordingly, the coolant passing through the power electronics coolant line **12** may pass through the battery coolant line **11**. As a result, the temperature of the coolant passing through the power electronics coolant line **12** may become similar to a temperature sensed by the temperature sensor **81** disposed in the battery coolant line **11**, and the controller **100** may determine the temperature of the coolant passing through the power electronics coolant line **12** based on the temperature sensed by the temperature sensor **81** disposed in the battery coolant line **11**.

(119) FIG. **8** illustrates a method of controlling the coolant system **10** in FIG. **2**.

(120) The controller **100** may determine whether the HVAC system **40** operates in the heating mode (S**11**).

(121) When it is determined that the HVAC system **40** operates in the heating mode, the controller **100** may determine whether the coolant system **10** operates in the third circulation mode (S**12**). In the third circulation mode, the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11**, and the radiator coolant line **13** may be fluidly blocked from the control valve **30**. That is, when the coolant system **10** operates in the third circulation mode, the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11** and the radiator coolant line **13**.

(122) When it is determined that the coolant system **10** operates in the third circulation mode, the controller **100** may determine whether a variation ΔT.sub.h in temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** is greater than or equal to a first predetermined threshold A (e.g., 2° C.) for a predetermined period of time (e.g., five seconds) (S**13**). While the vehicle is running at a constant speed, the temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** may be kept constant or the variation in the temperature of the heat sink **25***c* may be less than the predetermined threshold. While the vehicle is accelerating or decelerating, the vehicle speed may rapidly change or the torque of the electric motor **25***a* may rapidly change, and accordingly the variation in the temperature of the heat sink **25***c* sensed by the heat sink temperature sensor **29** may be greater than or equal to the predetermined threshold for a predetermined period of time.

(123) When it is determined that the variation ΔT.sub.h in the temperature of the heat sink **25***c* is greater than or equal to the first predetermined threshold A (e.g., 2° C.) for a predetermined period of time (e.g., five seconds), it may be determined whether a difference (T.sub.1−T.sub.h) between a temperature T.sub.1 of the coolant passing through the battery coolant line **11** and a temperature T.sub.h of the heat sink **25***c* is greater than or equal to a second predetermined threshold B (e.g., 10° C.) (S**14**). The second predetermined threshold B may be greater than the first predetermined threshold A.

(124) When it is determined that the difference (T.sub.1−T.sub.h) between the temperature T.sub.1 of the coolant passing through the battery coolant line **11** and the temperature T.sub.h of the heat sink **25***c* is greater than or equal to the second predetermined threshold B (e.g., 10° C.), the controller **100** may control the switching operation of the control valve **30** so that the coolant

system **10** may switch to the first circulation mode for a predetermined period of time (e.g., ten seconds) (S**15**). When the battery **21** is warmed up, the temperature of the coolant may relatively increase. Accordingly, when the difference (T.sub.1−T.sub.h) between the temperature T.sub.1 of the coolant passing through the battery coolant line **11** and the temperature T.sub.h of the heat sink **25***c* is greater than or equal to the second predetermined threshold B (e.g., 10° C.), thermal shock may occur as the coolant passes through the heat sink **25***c*. To prevent this, as the coolant system **10** operates in the first circulation mode, the power electronics coolant line **12** may be fluidly separated from the battery coolant line **11** and be fluidly connected to the radiator coolant line **13** so that the coolant having passed through the battery coolant line **11** may not be directed to the power electronics coolant line **12**. Here, the controller **100** may control the power electronic pump **16** in a manner that operates the power electronic pump **16** at maximum RPM. As the power electronic pump **16** operates at maximum RPM, thermal equilibrium may be quickly achieved between the temperature of the coolant passing through the power electronics coolant line **12** and the temperature of the coolant passing through the radiator coolant line **13**. In addition, the controller **100** may control the active air flap **18** in a manner that closes the active air flap **18**, and the controller **100** may control the cooling fan **19** in a manner that stops the cooling fan **19**. By closing the active air flap **18**, the front grille **17** may be fully closed. As the cooling fan **19** is stopped, the coolant passing through the radiator coolant line **13** may not be thermally affected by the ambient air. In a state in which the coolant system **10** switches to the first circulation mode for a predetermined period of time, the controller **100** may determine a temperature T of the coolant passing through the power electronics coolant line **12** based on a temperature sensed by the second temperature sensor **82** (S**16**).
(125) When it is determined that the difference (T.sub.1−T.sub.h) between the temperature T.sub.1 of the coolant passing through the battery coolant line **11** and the temperature T.sub.h of the heat sink **25***c* is less than the second predetermined threshold B (e.g., 10° C.), the controller **100** may control the switching operation of the control valve **30** so that the coolant system **10** may switch to the fourth circulation mode for a predetermined period of time (e.g., ten seconds) (S**17**). Here, the controller **100** may control the battery pump **15** and the power electronic pump **16** in a manner that operates the battery pump **15** and the power electronic pump **16** at maximum RPM. As the battery pump **15** and the power electronic pump **16** operate at maximum RPM, thermal equilibrium may be quickly achieved between the temperature of the coolant passing through the battery coolant line **11** and the temperature of the coolant passing through the power electronics coolant line **12**.
(126) In a state in which the coolant system **10** switches to the fourth circulation mode for a predetermined period of time, the controller **100** may determine the temperature T of the coolant passing through the power electronics coolant line **12** based on a temperature sensed by the first temperature sensor **81** (S**18**).
(127) When it is determined that the variation ΔT.sub.h in the temperature of the heat sink **25***c* is less than the first predetermined threshold A (e.g., 2° C.) for a predetermined period of time (e.g., five seconds), the controller **100** may determine the temperature T of the coolant passing through the power electronics coolant line **12** based on a temperature of the heat sink sensed by the heat sink temperature sensor **29** (S**19**).
(128) As set forth above, the vehicular coolant system and the method of controlling the same may be designed to have the temperature sensor provided in at least one of the battery coolant line, the power electronics coolant line, or the radiator coolant line, control the switching operation(s) of the control valve under predetermined condition(s) in a manner that fluidly connects a coolant line without any temperature sensor to a coolant line with the temperature sensor, and determine the temperature of the coolant passing through the coolant line without any temperature sensor based on the temperature sensed by the temperature sensor. By reducing the number of temperature sensors, cost savings may be achieved and electric efficiency may be improved.
(129) Hereinabove, although the present disclosure has been described with reference to exemplary

implementations and the accompanying drawings, the present disclosure is not limited thereto, but may be variously modified and altered by those skilled in the art to which the present disclosure pertains without departing from the spirit and scope of the present disclosure claimed in the following claims.

Claims

1. A coolant system, comprising: a battery coolant line fluidly connected to a battery; a power electronics coolant line fluidly connected to a power electronic component; a radiator coolant line fluidly connected to a radiator; a control valve configured to allow the battery coolant line, the power electronics coolant line, and the radiator coolant line to be fluidly connected to or separated from one another; a temperature sensor disposed at at least one of the battery coolant line, the power electronics coolant line, or the radiator coolant line; and a controller configured to control the control valve to: fluidly connect (i) a first coolant line that does not have the temperature sensor among the battery coolant line, the power electronics coolant line, and the radiator coolant line to (ii) a second coolant line that has the temperature sensor among the battery coolant line, the power electronics coolant line, and the radiator coolant line, and determine a first temperature of a coolant passing through the first coolant line based on a second temperature sensed by the temperature sensor disposed at the second coolant line.
2. The coolant system according to claim 1, wherein the temperature sensor is disposed at the battery coolant line.
3. The coolant system according to claim 1, wherein the temperature sensor comprises (i) a first temperature sensor disposed at the battery coolant line and (ii) a second temperature sensor disposed at the radiator coolant line.
4. The coolant system according to claim 1, further comprising a heat sink temperature sensor disposed at a heat sink of the power electronic component.
5. The coolant system according to claim 1, wherein the control valve comprises: a first battery-side port connected to an inlet of the battery coolant line; a second battery-side port connected to an outlet of the battery coolant line; a first power electronic-side port connected to an inlet of the power electronics coolant line; a second power electronic-side port connected to an outlet of the power electronics coolant line; a first radiator-side port connected to an inlet of the radiator coolant line; and a second radiator-side port connected to an outlet of the radiator coolant line.
6. The coolant system according to claim 5, wherein the control valve is configured to allow two or more of the first battery-side port, the second battery-side port, the first power electronic-side port, the second power electronic-side port, the first radiator-side port, or the second radiator-side port to selectively communicate with one another.
7. The coolant system according to claim 1, further comprising a battery pump fluidly connected to the battery coolant line.
8. The coolant system according to claim 1, further comprising a battery chiller that is fluidly connected to the battery coolant line and configured to thermally connect the battery coolant line to a heating, ventilation, and air conditioning (HVAC) system.
9. The coolant system according to claim 1, further comprising a power electronic pump fluidly connected to the power electronics coolant line.
10. The coolant system according to claim 1, further comprising a heat exchanger that is fluidly connected to the power electronics coolant line and configured to thermally connect the power electronics coolant line to an HVAC system.
11. A method for controlling a coolant system, the coolant system including a battery coolant line fluidly connected to a battery, a power electronics coolant line fluidly connected to a power electronic component, a radiator coolant line fluidly connected to a radiator, and a control valve configured to control flow of a coolant through the battery coolant line, the power electronics

coolant line, and the radiator coolant line, the method comprising: obtaining a temperature of a heat sink of the power electronic component; determining whether a variation of the temperature of the heat sink of the power electronic component is greater than or equal to a predetermined threshold in a first circulation mode in which the power electronics coolant line is fluidly separated from the battery coolant line; based on determining that the variation of the temperature of the heat sink of the power electronic component is greater than or equal to the predetermined threshold, switching to a second circulation mode in which the power electronics coolant line is fluidly connected to the battery coolant line; obtaining a first temperature sensed by a temperature sensor that is disposed at the battery coolant line; and determining a second temperature of the coolant passing through the power electronics coolant line based on the first temperature sensed by the temperature sensor disposed at the battery coolant line.

12. The method according to claim 11, further comprising: based on determining that the variation of the temperature of the heat sink of the power electronic component is less than the predetermined threshold, obtaining a third temperature that is sensed by a heat sink temperature sensor; and determining the second temperature of the coolant passing through the power electronics coolant line based on the third temperature sensed by the heat sink temperature sensor.

13. A method for controlling a coolant system, the coolant system including a battery coolant line fluidly connected to a battery, a power electronics coolant line fluidly connected to a power electronic component, a radiator coolant line fluidly connected to a radiator, and a control valve configured to control flow of a coolant through the battery coolant line, the power electronics coolant line, and the radiator coolant line, the method comprising: obtaining a temperature of a heat sink of the power electronic component; determining whether a variation of the temperature of the heat sink of the power electronic component is greater than or equal to a first predetermined threshold in a first circulation mode in which the power electronics coolant line is fluidly separated from the battery coolant line and the radiator coolant line; based on determining that the variation of the temperature of the heat sink of the power electronic component is greater than or equal to the first predetermined threshold, determining a temperature difference between a first temperature of the coolant passing through the battery coolant line and the temperature of the heat sink; determining whether the temperature difference is greater than or equal to a second predetermined threshold; based on determining that the temperature difference is greater than or equal to the second predetermined threshold, switching to a second circulation mode in which the power electronics coolant line is fluidly connected to the radiator coolant line; and obtaining a first sensor temperature sensed by a first temperature sensor disposed at the radiator coolant line; and determining a second temperature of the coolant passing through the power electronics coolant line based on the first sensor temperature sensed by the first temperature sensor disposed at the radiator coolant line.

14. The method according to claim 13, further comprising: obtaining a second sensor temperature sensed by a heat sink temperature sensor based on determining that the variation of the temperature of the heat sink of the power electronic component is less than the first predetermined threshold; determining the second temperature of the coolant passing through the power electronics coolant line based on the second sensor temperature sensed by the heat sink temperature sensor.

15. The method according to claim 13, further comprising: switching to a third circulation mode in which the power electronics coolant line is fluidly connected to the battery coolant line based on determining that the temperature difference is less than the second predetermined threshold; obtaining a second sensor temperature sensed by a second temperature sensor disposed at the battery coolant line; and determining the second temperature of the coolant passing through the power electronics coolant line based on the second sensor temperature sensed by the second temperature sensor disposed at the battery coolant line.